How Predictable Are Michelin Star Rankings?

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Background:
We investigated whether Michelin stars are efficiently allocated across global markets in 2026 given economic, geographic, and market constraints. The analysis proceeds from raw web-scraped restaurant data through negative binomial regression to a country-level Star Allocation Residual: a measure of how much each country's star count departs from what its economic and market characteristics would predict.

Six primary data sources were used. URLs are provided in the table below.

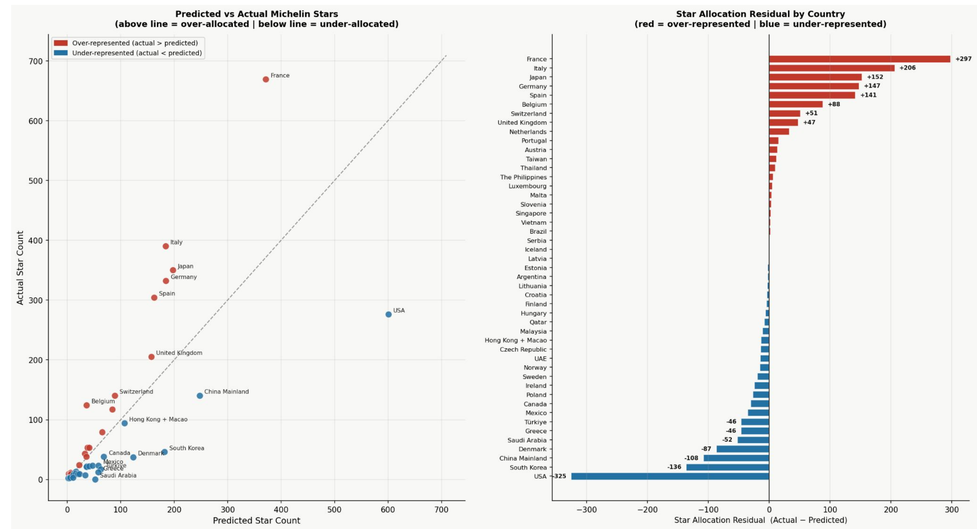
Dataset	Source	URL	Coverage
Michelin Guide 2026	Algolia API (guide.michelin.com)	guide.michelin.com/us/en/restaurants…	Scraped March 2026
GDP per Capita	World Bank WDI — NY.GDP.PCAP.CD	data.worldbank.org/indicator/NY.GDP.PC…	2022–2024
GDP per Capita (Taiwan)	IMF World Economic Outlook	www.imf.org/external/datamapper/NGDPDP…	2022–2024
Urbanization Rate	World Bank WDI — SP.URB.TOTL.IN.ZS	data.worldbank.org/indicator/SP.URB.TO…	2022–2024
Tourism Arrivals	UNWTO World Tourism Barometer	www.unwto.org/tourism-data/global-and-…	2022–2024
Restaurant Counts	Eurostat Structural Business Statistics (NACE 56.1)	ec.europa.eu/eurostat/statistics-expla…	Most recent year

Methodology:

Data Collection: The 2026 Michelin Guide was scraped in March 2026 via the Algolia API, yielding 18,837 restaurants across 53 countries. Four external economic variables were compiled for each of 47 model countries.

Data Cleaning: Several territories were consolidated for modelling and Michelin's local-currency pricing tiers were converted to USD using March 2026 exchange rates.

Analytical Method: Restaurant-level data were collapsed to 47 country rows. A Negative Binomial regression (NB2) was fitted with total starred restaurants as the dependent variable and our external economic variables as the independents. The Star Allocation Residual — actual minus predicted stars — was computed for each country to identify systematically over- and under-represented markets.



Over-Represented ▲				Under-Represented ▼			
Country	Actual	Predicted	Residual	Country	Actual	Predicted	Residual
France	669	371.6	+297.4	USA	276	601.2	−325.2
Italy	390	184.4	+205.6	South Korea	46	182.2	−136.2
Japan	350	197.9	+152.1	China Mainland	140	248.1	−108.1
Germany	332	184.7	+147.3	Denmark	37	123.7	−86.7
Spain	304	162.8	+141.2	Türkiye	17	63.0	−46.0
Belgium	124	36.3	+87.7	Greece	12	58.3	−46.3
Switzerland	140	89.3	+50.7	Saudi Arabia	0	52.5	−52.5
United Kingdom	205	157.6	+47.4	Canada	38	68.5	−30.5

Initial Takeaways:

The model confirms that Michelin star allocation is meaningfully but incompletely explained by structural variables, which are all highly significant ($p < 0.001$) with similar-sized coefficients, meaning no single factor dominates.

The dispersion parameter (alpha = 0.52) tells us there's substantial unexplained variance left over. Implying that something beyond economics is driving Michelin's decisions.

The residuals of the regression demonstrate where structural factors fail to explain the amount of Michelin Stars awarded. Namely in the most represented country France (+297), and the least represented, the United States (−325).

Conclusion:

We determined that Michelin Star rankings, while highly correlated with strong economies, are not completely explained by them. Especially with regard to outliers such as France and the United States. Further analysis will examine other notable explanatory variables, such as type of cuisine, exact city, and per restaurant pricing. Generative classification and decision trees will likely be used for this segment of analysis.